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Comprehensive assessment of hydrogen injection in natural gas networks: Using dimensional analysis and reduced-order models for mixture quality prediction

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ABSTRACT

This study presents a physics-based reduced-order model (ROM) to assess the mixing behavior of hydrogen and natural gas in pipelines, facilitating the integration of hydrogen into natural gas networks—a key strategy for reducing carbon emissions. The methodology consists of dimensional analysis to identify crucial dimensionless parameters, CFD simulations for data collection, and statistical analysis for effective data modeling. This approach enables rapid evaluation of injection strategies and operational conditions, reducing the computational demands of traditional CFD simulations. The model correlates reduced-order parameters with key dimensionless numbers like Reynolds number, and diameter and velocity ratios, to develop a predictive tool for assessing mixture quality. Validated through extensive simulations, the model accurately predicts gas mixing, using beta distributions for statistical representation to enhance data interpretability and applicability.

1. Introduction

Green hydrogen technologies have emerged as pivotal components in the contemporary energy landscape, spurred by the pressing need to combat climate change and curtail greenhouse gas emissions. Green hydrogen, with its climate neutrality, is a promising energy vector [1]. It serves as an energy carrier for renewable energies and a solution for the storage of renewable energy [2]. Given these properties, the transport and distribution of hydrogen are widely researched and studied to achieve an optimal integration of this resource into the energy mix [3–5].

The most promising option for hydrogen distribution is the use of existing natural gas infrastructure, which is considered the most feasible solution in the short and medium term from both technical and economic perspectives [6,7]. Transportation of natural gas enriched with hydrogen is currently the most efficient and cost-effective method [8]. However, the mixture of these two gases poses various challenges due to their different thermodynamic properties. These challenges include concerns related to the integrity of pipe materials, safety issues, leaks, and the need to modify pipelines, storage tanks, compressors, and end-user devices [2,8–12]. Therefore, understanding the intricacies of hydrogen-natural gas mixtures is paramount, not only for safety considerations but also to optimize operational efficiency [13].

A key process to mitigate the potential hazards that arise from the combined transportation of hydrogen and natural gas is the injection of hydrogen into natural gas pipelines [14]. Achieving a homogeneous mixture of hydrogen and natural gas in the shortest time is imperative to prevent the mentioned issues. Inadequate mixing of these gases can engender multifaceted risks: (i) elevated concentrations of hydrogen may catalyze degradation in the metallic infrastructure of natural gas pipelines [15–17]; and (ii) sub-optimal mixing can result in end-users receiving a blend with a higher-than-anticipated hydrogen content, potentially leading to operational challenges in utilizing the fuel [18–20].

For this reason, hydrogen injection into natural gas pipelines has been the subject of extensive research in scientific literature, using both experimental methods and computational fluid dynamics (CFD) to optimize the mixing process and avoid the potential issues indicated above. One example is the work of Zhou et al. [21], who developed dynamic models by numerically solving 1D partial differential equations to simulate hydrogen injections in gas networks. Liu et al. [22] explored mixing dynamics using both experimental setups and CFD in a static mixer, offering a dual perspective on the phenomenon. Yan et al. [23] analyzed the influence of the hydrogen inlet configurations on mixing efficiency through CFD simulations. This work highlighted the critical role of structural variations in pipeline design. Concurrently, Su

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et al. [24] utilized CFD to assess the quality of hydrogen-natural gas mixtures under various injection conditions. Furthermore, in this work a neural network model is developed to predict variations in H_2 concentration. Significant contributions also come from studies such as Eames et al. [14] and Khabbazi et al. [8], which examined the effects of T-junction orientation on mixing efficiency, and Smith et al. [25], which compared different turbulence models to determine the most accurate for predicting fluid behavior in T-junction configurations. Additionally, the dynamics of mixing in T-junctions, whether involving gases or hot and cold water, have been extensively studied, employing both experimental approaches and CFD techniques [26–29].

This article presents a novel approach to design hydrogen injections into natural gas networks. The traditional approach consists of applying numerical fluid dynamic techniques (CFD) to design and optimize it. One of the main inconveniences of this approach is the time and cost required, as CFD techniques take several hours or even days to simulate a use case. The approach proposed involves creating a reduced physics model that can replicate the results of detailed numerical simulations in real-time. To achieve this, the authors have developed a methodology that combines traditional computational fluid dynamics (CFD) simulations with specific techniques of physics-based reduced-order models like dimensional and statistical analysis. This method produces a real-time model that can compute the hydrogen-natural gas mixture for a predefined geometric configuration, such as a T-junction. The model considers parameters like the flow ratio between the natural gas network and the hydrogen injection or the diameter ratio between the respective injection pipelines.

While physics-based reduced-order models have been developed for other sectors, this approach is novel for H_2 injection. As a result, the outcome (a real-time model to simulate the H_2 injection) is particularly innovative. It has the potential to significantly reduce time and computational costs for the design of hydrogen injections into natural gas pipelines. Since the industry tends to develop standard solutions in which only a few parameters are allowed to vary, this model can be applied massively and impact the sector significantly.

The integration of CFD and reduced-order models (ROMs) has become a powerful approach in engineering research to address the computational intensity of detailed simulations. This method involves using CFD simulations to generate detailed data sets, which then inform the development of more computationally efficient ROMs through various techniques such as artificial intelligence, statistical methods, or simplified physical principles. For instance, König-Haagen et al. [30] successfully apply this strategy to develop ROMs for latent heat thermal energy storage systems, using data derived from CFD simulations. Similarly, Yang et al. [31] utilize proper orthogonal decomposition and Galerkin projection to create a ROM of a lead–bismuth reactor, demonstrating the effectiveness of this approach in simplifying complex thermal–hydraulic behaviors. Additionally, Li et al. [32] employ dimensional analysis in conjunction with CFD results to construct a predictive model for hydrogen dispersion in leakage scenarios, showcasing the versatility of ROMs in various application contexts.

Further expanding ROM techniques, Zambrano et al. [33] explore tensor factorization for building surrogate models from sparse data, presenting an alternative to Gaussian process-based methods which often lack interpretability. This approach, implemented in the Twinkle library, effectively handles sparse data from CFD and FEM applications and is useful in both academic and industrial settings, especially for data analysis and post-processing. Using this library, García-Camprubí et al. [34] develop real-time virtual sensors for microfluidic processes, illustrating how ROMs can improve control systems in manufacturing, such as factory-on-chips, by predicting fluid dynamics that are difficult to measure directly.

These examples highlight the growing trend of leveraging high-fidelity CFD simulations to underpin simpler, yet robust models that can significantly reduce computational demands while maintaining accuracy.

The following section describes the methodology used to create the physics-based reduced-order model. After that, the results obtained are presented, and finally, we include the conclusions drawn from this work.

2. Methodology and setup

As case study, our research employs a straightforward T-junction setup for hydrogen injection at the bottom of the primary pipeline, a configuration that has been extensively studied and validated in previous research. Eames et al. [14] and Khabbazi et al. [8] specifically demonstrated that lower injection points significantly enhance mixing by leveraging gravitational effects, compared to upper T-junction configurations which often result in poor mixing due to stratification. This result supported our decision to use the bottom injection approach, as it has been proven to optimize the mixing process. In this context, Su et al. [24] further explored the dynamics of hydrogen injection using CFD simulations to fine-tune the mixing efficiency.

The objective of the methodology developed is to obtain a physics-based reduced-order model that can calculate the degree of mixing of hydrogen and natural gas in different sections downstream the injection. The methodology consists of three stages: dimensional analysis, CFD simulation, and assessment of the mixture degree.

Dimensional analysis allows the determination of dimensionless variables on which the degree of mixing depends, reducing the problem's dimensionality. This results in a considerable reduction of the CFD simulations required to obtain the dataset necessary to characterize the hydrogen injection.

The second stage consists of the configuration and execution of CFD simulations of the hydrogen injection into natural gas pipelines. The results of these simulations are processed to obtain the dataset that will feed the reduced-order model.

The third stage consists of the elaboration of the reduced-order model and the adjustment of the mathematical expression that determines the hydrogen degree of mixing downstream the injection.

2.1. Dimensional analysis

A comprehensive exploration of all possible variable combinations involved in the injection process would require a huge amount of simulations, entailing significant computational resources and complicating the analysis of results. To mitigate these challenges, we have employed dimensional analysis to reduce the dimensionality of the problem. This approach serves as a cornerstone in the development of our reduced-order model by identifying key dimensionless parameters that condense the fundamental dynamics of the mixing process. This reduction not only aids in generalizing the solution across a wide range of conditions but also enhances the interpretability and applicability of the model.

First, we introduce a dimensionless dependent variable, denoted as M, which characterizes the degree of mixture in specific pipeline sections downstream the junction. While this variable will be defined in detail in the next section, we use it here generically to outline our methodological approach.

The independent variables were initially defined as pipe diameters (d_{CH_4}, d_{H_2}), pressure (P), temperature (T), methane velocity (v_{CH_4}), and mixture composition ($\%H_2$):

$$M = f(d_{CH_4}, d_{H_2}, P, T, v_{CH_4}, \%H_2) \tag{1}$$

For mathematical convenience and to focus on the fundamental properties affecting fluid dynamics, we redefined the variables in terms of mixture gas density (ρ), mixture gas viscosity (μ_g), and methane and hydrogen velocities (v_{CH_4}, v_{H_2}):

$$M = f'(d_{CH_4}, d_{H_2}, \rho, v_{CH_4}, v_{H_2}, \mu_g) \tag{2}$$

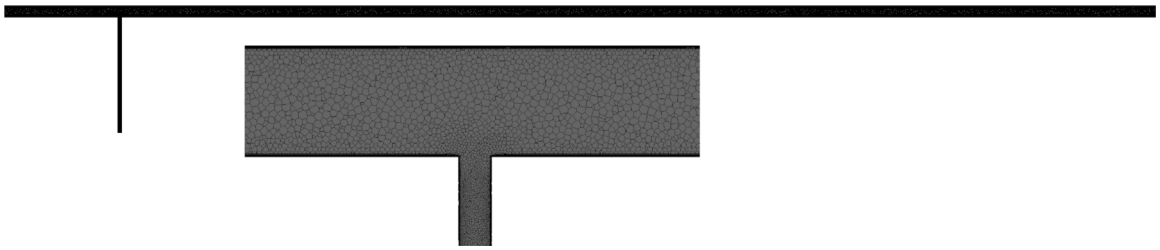

Fig. 1. Representative geometry and mesh of the T-junction setup used in this study.

Subsequently, we derived a dimensionless representation through dimensional analysis:

$$M = F\left(\frac{d_{H_2}}{d_{CH_4}}, \frac{v_{H_2}}{v_{CH_4}}, Re\right) \tag{3}$$

where Re denotes the Reynolds number, a fundamental dimensionless quantity in fluid mechanics, offering a well-established and interpretable measure of fluid flow characteristics.

The choice of these particular dimensionless ratios (diameter and velocity ratios) provides clear insights into the mixing dynamics, thereby simplifying the complexity of the full-scale computational model into more tractable terms.

This methodological approach enables an efficient assessment of the mixture quality across various parameter combinations. Using dimensional analysis, we mathematically reduced the number of variables from 6 dimensional ones to 3 dimensionless ones. Despite this reduction, no physical information is lost in the process. This reduction significantly decreased the number of CFD simulations required to obtain the necessary dataset for characterizing hydrogen injection. Thus, by running a reduced number of simulations, primarily varying the Reynolds number, d_{H_2}/d_{CH_4}, and v_{H_2}/v_{CH_4}, we gained insights into how these parameters influence mixing behavior within the pipeline.

2.2. CFD setup

The CFD simulation setup follows to the methodologies described by Su et al. [24] and Eames et al. [14], which we adopt to generate the synthetic data necessary for developing our ROM. The Reynolds-averaged Navier–Stokes (RANS) equations are solved for continuity, momentum, energy, and species (H_2), assuming an ideal gas model. Turbulence within the system is modeled using the k-ω SST model [35], selected for its proven accuracy in predicting flow behavior in complex geometries similar to our T-junction setup. Previous studies, such as Smith et al. [25], have highlighted the efficacy of this model in capturing essential dynamics in gas mixing scenarios, where phenomena like buoyancy play a significant role.

While we used the described CFD setup, the methodology for ROM generation presented in this work can be applied to any CFD setup, allowing scalability with more advanced models (such as LES) as computational resources permit.

The computational domain consists of a main pipe with a length of 10 m, with the T-junction placed ten diameters downstream the CH_4 entrance. The mesh includes 7.5 to 12 million polyhedral cells, varying with configuration, and it is refined near the walls to ensure that the wall-spacing parameter y^+ remains approximately 1, reflecting a practice aligned with standards from referenced studies to ensure accuracy without conducting a separate mesh independence study. For better clarity of the T-junction setup and the injection scenarios tested in this work, a representative geometry and mesh are shown in Fig. 1.

This investigation encompasses six distinct geometries, characterized by the main pipe diameter, d_1, ranging from 4 to 15 cm, and the hydrogen inlet pipe diameter, d_2, varying from 1 to 3 cm. These variations result in d_2/d_1 ratios spanning from 0.1 to 0.6. Note that extreme cases such as 1/15 or 3/4 were not simulated; instead, the actual simulated extremes were 1/10 and 3/5. Operational parameters were diversified to analyze a comprehensive range of scenarios. The velocity of natural gas, v_{CH4}, varied from 1 to 10 m/s, and the ratio of hydrogen to natural gas velocities, v_{H_2}/v_{CH_4}, ranged from 2 to 150. The operating pressure fluctuated between 5 and 20 bar, with a consistent temperature of 27 °C. These conditions generated Reynolds numbers ranging from 7.5×10^3 to 1.18×10^6, covering a spectrum of flow conditions for a thorough investigation. In total, 86 simulations were conducted, providing comprehensive coverage of the parameter space defined by the three dimensionless numbers illustrated in Fig. 2.

2.3. Reduced-order model of mixture degree

Defining a "good" mixture is inherently subjective and can vary based on the specific application or system conditions. To establish an objective metric, we quantified the degree of mixture based on the mass fraction of H_2 within the pipeline. The challenge then becomes how to best represent these mass fraction statistically.

While computational fluid dynamics (CFD) allows for the calculation and visualization of mass fraction contours (Fig. 3), providing detailed spatial distributions, this approach often results in complex datasets. Although visualizing these contours offers comprehensive insights into the mixing process, comparing them across different cases can be cumbersome due to their complexity.

To simplify the analysis, we moved from visualizing detailed contours to summarizing the data with histograms of mass fractions at various pipeline sections. These sections are defined by the dimensionless ratio L/d_{CH_4}, where L is the distance from the junction. This method not only makes the information easier to analyze but also helps compare mixing under different operational conditions. Fig. 4 displays the histograms of mass fractions, showing the distribution of data points.

Next, we computed Probability Density Functions (PDFs) from these histograms. We fit these PDFs to beta distribution functions for each section across all simulated cases. The beta distribution is particularly appropriate because it ranges from 0 to 1, matching the nature of mass fractions, and it can flexibly model the distribution shapes we observe in the CFD simulations. Fig. 5 shows the adjustment to a beta distribution, compared to the PDF computed using Kernel Density Estimation (KDE) [36], which smooths the discrete data points of the histogram into a continuous probability distribution, allowing us to directly assess the fit's accuracy.

The beta distribution can be parameterized using just two parameters, typically denoted as α and β. Alternatively, it can be expressed in terms of μ (mean) and ν (sample size), which provide intuitively interpretable physical meanings. This parameterization helps in simplifying the representation of complex data and facilitates easier analysis and comparison of mixture quality across different conditions. The hydrogen concentrations in each section were well represented by the beta distribution (see Fig. 5). Therefore, the next step involves obtaining a correlation based on previously defined dimensionless numbers to calculate the beta function parameters, μ and ν. By obtaining a good correlation, we will be able to determine the mixture in the defined sections of the pipe downstream the injection.

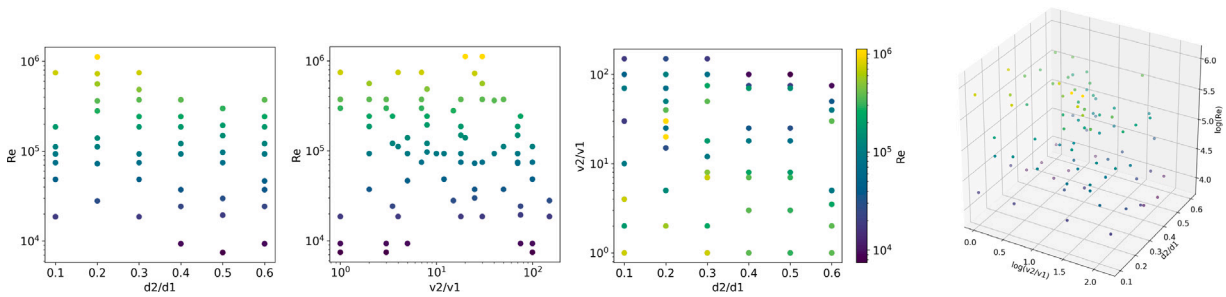

Fig. 2. Visualization of the parameter space exploration, where each point corresponds to a simulation and is characterized by the values of the three dimensionless numbers: d_{H_2}/d_{CH_4}, v_{H_2}/v_{CH_4}, and Re. Dots colored with Reynolds number. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

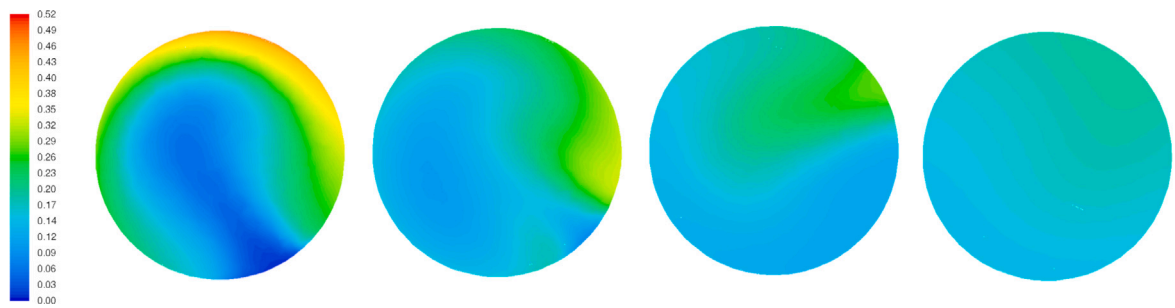

Fig. 3. Contours of H_2 mass fraction in sections of a pipe for L/d_{CH_4} equal to 2, 5, 10 and 50 (from left to right).

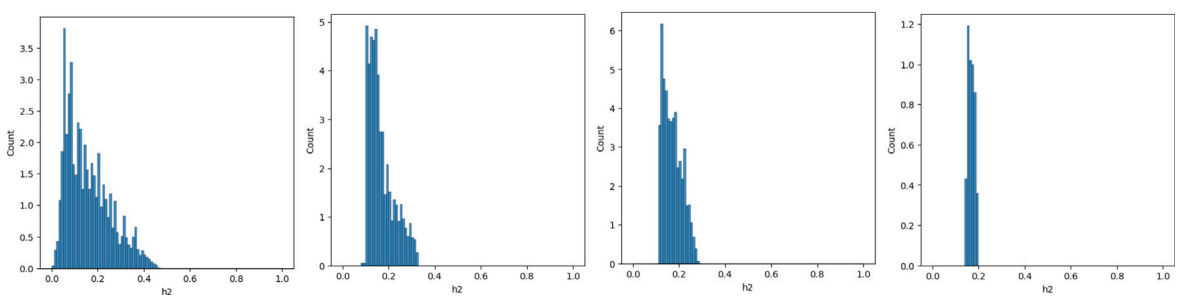

Fig. 4. Histogram of H_2 mass fraction (weighted with cross area) in sections of a pipe for L/d_{CH_4} equal to 2, 5, 10 and 50 (from left to right).

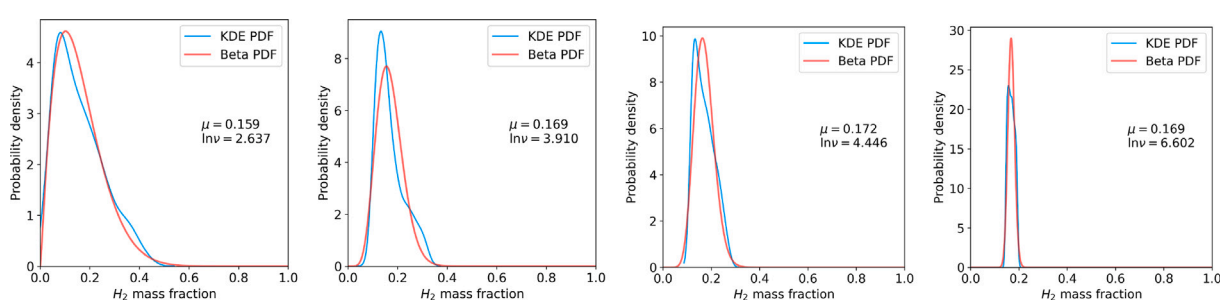

Fig. 5. PDF and beta approximation for H_2 mass fraction in sections of a pipe for L/d_{CH_4} equal to 2, 5, 10 and 50 (from left to right).

The correlation established for the parameters μ and $\ln(\nu)$ takes the form of the mathematical expression defined in Eq. (4). Here, ξ represents the general form used to fit both μ and $\ln(\nu)$. This means that ξ can be replaced by either μ or $\ln(\nu)$ depending on the context, and the specific set of constants A_1 to D_6 will vary accordingly. Thus, these correlations only depend on the three dimensionless numbers determined for the problem ($N_d = d_{H_2}/d_{CH_4}$, $N_v = v_{H_2}/v_{CH_4}$, and Re). Given the disparity of absolute values for these numbers, for mathematical convenience, we use normalized values, defined as $\hat{\phi} = 1 + (\phi - \phi_{min})/(\phi_{max} - \phi_{min})$. The maximum and minimum values used for the normalization for each dimensionless number are defined in Table 1. The form of this expression was chosen because it is composed of simple terms, allowing us to analyze the specific influence of each variable. We tested different expressions of this type and found that this form provided a suitable balance between the number of terms and accuracy. While a simpler expression might suffice for μ, we used the same form for both parameters for consistency.

The fitting of the parameters A_1 to D_6 in Eq. (4) was performed using the `curve_fit` function from the `scipy.optimize` library [37]. This function employs non-linear least squares to fit the

Table 1
Minimum and maximum values of dimensionless numbers used for normalization.

	ϕ_{min}	ϕ_{max}
N_d	0.1	0.6
N_v	1.0	150
Re	7459	1188750

model to the data. Specifically, `curve_fit` adjusts the parameters to minimize the difference between the predicted values (from the mathematical expression) and the actual CFD results. This optimization process iteratively updates the parameters to find the best fit that reduces the error, ensuring that the model accurately represents the underlying data.

While methods like neural networks might provide more accurate results with reasonable computational efficiency, this approach uses an algebraic expression that offers advantages in terms of ease of implementation. This model can be seamlessly integrated into various applications, including spreadsheets and other simple computational tools, making it highly accessible for engineers. The algebraic nature of the model allows for interpretation and quick calculations, facilitating real-time decision-making without the need for specialized software or extensive computational resources.

$$\begin{aligned}\xi =& A_1 \ln(\widehat{N_v}) + A_2 \ln(\widehat{Re}) + A_3 \ln(\widehat{N_d}) + \\ & A_4 \widehat{N_v} + A_5 \widehat{Re} + A_6 \widehat{N_d} + \\ & A_7 \widehat{N_v}^2 + A_8 \widehat{Re}^2 + A_9 \widehat{N_d}^2 + \\ & A_{10} \widehat{N_v}^{-1} + A_{11} \widehat{Re}^{-1} + A_{12} \widehat{N_d}^{-1} + \\ & B_1 \widehat{N_v} \widehat{Re} + B_2 \widehat{N_v} \widehat{N_d} + B_3 \widehat{Re} \widehat{N_d} + \\ & B_4 \widehat{N_v}^2 \widehat{Re} + B_5 \widehat{N_v}^2 \widehat{N_d} + B_6 \widehat{Re}^2 \widehat{N_v} + \\ & B_7 \widehat{Re}^2 \widehat{N_d} + B_8 \widehat{N_d}^2 \widehat{N_v} + B_9 \widehat{N_d}^2 \widehat{Re} + \\ & C_1 \widehat{N_v} \ln(\widehat{N_d}) + C_2 \widehat{Re} \ln(\widehat{N_d}) + C_3 \widehat{N_d} \ln(\widehat{N_v}) + \\ & C_4 \widehat{Re} \ln(\widehat{N_v}) + C_5 \widehat{Re} \ln(\widehat{Re}) + C_6 \widehat{N_d} \ln(\widehat{Re}) + \\ & D_1 \widehat{Re} \exp(\widehat{N_v}) + D_2 \widehat{N_v} \exp(\widehat{Re}) + D_3 \widehat{N_v} \exp(\widehat{N_d}) + \\ & D_4 \widehat{N_d} \exp(\widehat{N_v}) + D_5 \widehat{N_d} \exp(\widehat{Re}) + D_6 \widehat{Re} \exp(\widehat{N_d})\end{aligned} \tag{4}$$

3. Results

This section details the results obtained from the comprehensive application of the developed methodology over 86 simulation cases, encompassing a wide range of operational scenarios. The primary outcome of this study is the establishment of a robust correlation model that predicts the mixture quality using the parameters μ and $\ln(\nu)$, derived from dimensionless numbers.

3.1. Parameter optimization

Tables 2 and 3 present the optimized values for the parameters A_1 to D_6 in the correlation proposed for μ and $\ln(\nu)$. The values are detailed for six dimensionless distances, L/d_{CH_4}, downstream the T-junction, providing calculations that predict the evolution of the mixture up to a length equivalent to 50 diameters from the mixing point.

3.2. Validation of the correlation

To assess the accuracy of our correlation, validation plots compare the true values with the predicted values for μ and $\ln(\nu)$. Fig. 6 displays these plots, where different dot colors represent various values of the dimensionless length L/d_{CH_4}. The correlation coefficient R^2 for μ is 0.997, indicating an excellent fit, while for $\ln(\nu)$, it is 0.945. As L/d_{CH_4} increases, the values of ν tend to rise, consistent with the inverse relationship to variance, reflecting an increase in mixture uniformity.

Table 2
Values of parameters A_1 to D_6 for the correlation equation for μ for different L/d_{CH_4} values.

L/d_{CH_4}	1	2	5	10	20	50
A_1	−1.96E+3	−1.02E+3	−3.91E+2	−3.36E+2	−3.20E+2	−3.56E+2
A_2	3.84E+2	2.86E+2	1.51E+2	1.97E+2	2.24E+2	2.28E+2
A_3	−1.60E+2	−1.82E+2	−1.13E+2	−6.79E+1	−7.74E+1	−6.05E+1
A_4	1.57E+3	8.36E+2	3.79E+2	3.22E+2	2.97E+2	3.22E+2
A_5	−1.09E+3	−6.73E+2	−3.56E+2	−3.50E+2	−3.59E+2	−3.75E+2
A_6	2.12E+2	2.51E+2	1.94E+2	1.80E+2	1.97E+2	1.92E+2
A_7	−2.24E+2	−1.29E+2	−6.16E+1	−5.33E+1	−5.07E+1	−5.41E+1
A_8	−9.01E+1	−5.18E+1	−3.23E+1	−1.51E+1	−8.28E+0	−8.32E+0
A_9	−8.31E+0	−1.12E+1	−6.59E+0	−3.77E+0	−4.21E+0	−2.60E+0
A_{10}	−7.75E+2	−3.79E+2	−1.21E+2	−1.03E+2	−9.62E+1	−1.11E+2
A_{11}	5.89E+1	5.80E+1	3.16E+1	5.74E+1	7.11E+1	7.19E+1
A_{12}	−8.54E+1	−9.18E+1	−5.77E+1	−3.42E+1	−3.94E+1	−3.21E+1
B_1	4.70E+2	2.41E+2	9.64E+1	1.03E+2	1.06E+2	1.24E+2
B_2	−8.80E+1	−8.94E+1	−1.25E+2	−1.23E+2	−1.17E+2	−1.23E+2
B_3	4.29E+1	2.06E+1	3.66E+1	−6.29E−1	−1.51E+1	−2.16E+1
B_4	−2.23E+2	−1.26E+2	−5.90E+1	−5.73E+1	−5.70E+1	−6.34E+1
B_5	5.46E+1	5.30E+1	4.54E+1	4.56E+1	4.63E+1	4.87E+1
B_6	1.15E+1	8.36E+0	8.00E+0	8.53E+0	6.29E+0	4.65E+0
B_7	−2.64E+1	−1.30E+1	−1.03E+1	−3.53E+0	2.16E+0	3.60E+0
B_8	−2.62E+1	−2.42E+1	−5.89E+0	−7.80E+0	−1.07E+1	−1.13E+1
B_9	−2.81E+0	−3.59E+0	−1.14E+1	−2.90E+0	−7.35E−1	1.29E+0
C_1	−4.62E+1	−4.30E+1	−1.29E+1	−1.48E+1	−1.97E+1	−2.07E+1
C_2	−1.54E+1	−1.21E+1	−2.18E+1	−6.16E+0	−3.26E+0	−4.06E−1
C_3	9.95E+1	9.90E+1	9.14E+1	9.33E+1	9.49E+1	9.92E+1
C_4	−2.54E+2	−1.31E+2	−5.41E+1	−5.95E+1	−5.97E+1	−6.79E+1
C_5	4.53E+2	2.79E+2	1.56E+2	1.16E+2	1.03E+2	1.07E+2
C_6	−4.97E+0	2.60E+0	3.31E+0	8.16E+0	1.22E+1	1.24E+1
D_1	8.05E+1	4.86E+1	2.49E+1	2.30E+1	2.25E+1	2.44E+1
D_2	−5.06E+0	−3.61E+0	−3.48E+0	−3.78E+0	−2.78E+0	−2.09E+0
D_3	6.44E+0	5.77E+0	7.94E−1	1.45E+0	2.21E+0	2.35E+0
D_4	−1.36E+1	−1.32E+1	−1.09E+1	−1.09E+1	−1.10E+1	−1.16E+1
D_5	1.36E+1	7.72E+0	6.38E+0	3.49E+0	1.12E+0	4.25E−1
D_6	−3.98E−1	3.58E−1	2.84E+0	7.50E−1	7.19E−2	−5.37E−1

Interestingly, the dispersion of $\ln(\nu)$ values also increases with higher L/d_{CH_4}, ranging from an R^2 of 0.97 at $L/d_{CH_4} = 1$ to 0.82 at $L/d_{CH_4} = 50$. This pattern suggests more nuanced dynamics as the gas travels further from the injection point. One possible explanation for this behavior is the exponentially higher values of ν, which suggest a distribution increasingly resembling a delta function. This implies greater uncertainty in the method at these points, potentially due to the higher homogeneity of the mixture, which paradoxically makes the precise characterization of mixture quality less critical at these stages due to better overall mixing. Such observations highlight the complex interplay between mixing efficiency and measurement certainty, underscoring the need for careful interpretation of model predictions in varied operational contexts.

3.3. Interpretation of correlation results

Fig. 7 offers a colormap representation of the correlation results (μ and $\ln(\nu)$) in a 2D space defined by the dimensionless numbers $N_d = d_{H_2}/d_{CH_4}$ and $N_v = v_{H_2}/v_{CH_4}$. The visual analysis shows smooth transitions in the function values across the parameter space, indicating a stable and consistent model behavior under varying conditions. These smooth gradients, while visually simplistic, underscore the capacity of the model to handle complex real-world scenarios without resorting to overly intricate or erratic solutions. Such stability in the model's predictions across diverse conditions is a strong indicator of its robustness and reliability.

To aid in interpreting the Figure, we included iso-lines of constant hydrogen blend ratio (HBR, a commonly used metric in this context [17,24,38–42]) values. The relationship between HBR and the dimensionless numbers is given by $N_d^2 N_v = \mathrm{HBR}/(1-\mathrm{HBR})$. Including these iso-lines helps in visualizing how the dimensionless parameters relate to practical HBR values, enhancing the interpretability of the figure.

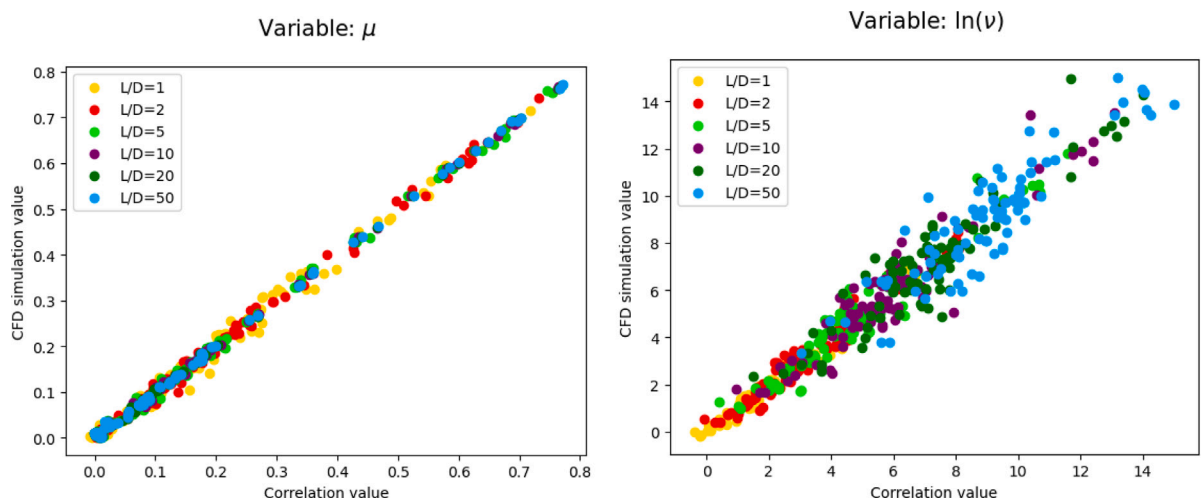

Fig. 6. Validation plots comparing true values with correlated values for μ (left) and $\ln(\nu)$ (right). Different dot colors represent various values of the dimensionless length L/d_{CH_4}. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

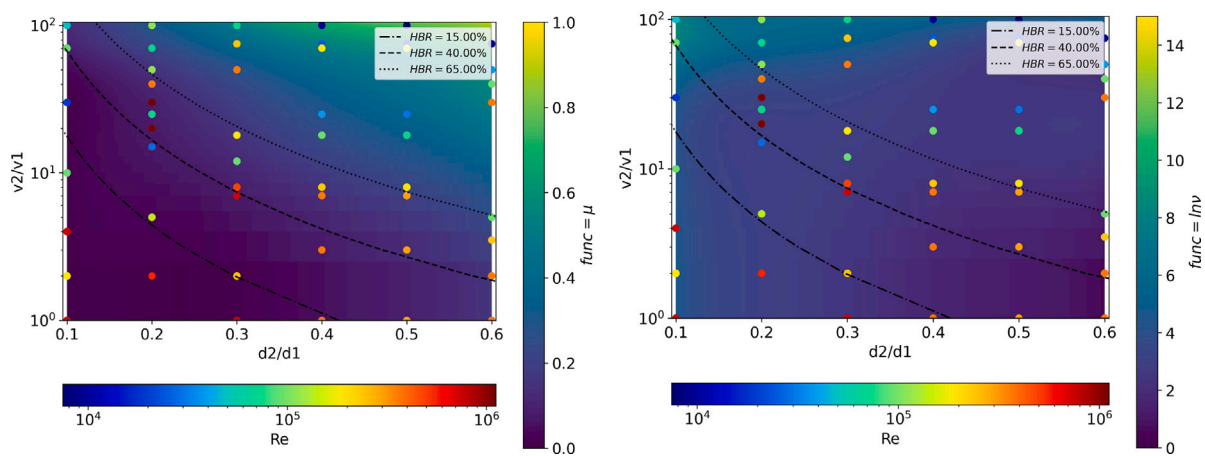

Fig. 7. Contours of correlated values of μ (left) and $\ln(\nu)$ (right) for constant Re=10^5 and $L/d_{CH_4} = 2$, for the range studied of N_d and N_υ. The dots (colored with Reynolds number) indicate the simulations in the phase space. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. Computational details

The CFD simulations were performed using Ansys Fluent. Each case required approximately 100 CPU-hours, consumed around 20 GB of RAM and used 2 Gb of storage for each case. The simulations were executed on an AMD Ryzen 5900X processor.

The post-processing of CFD-generated data, including the generation of histograms, PDFs, fitting to beta distribution functions, and parameter optimization for the tables Tables 2 and 3, took about 10 min of computational time.

The evaluation of the parameters using the algebraic expression and the reconstruction of the beta function are nearly instantaneous, demonstrating the efficiency of the ROM approach compared to the full CFD simulations.

5. Conclusions

This study has developed and validated a novel reduced-order model to assess the mixing quality of hydrogen in natural gas pipelines, utilizing dimensional analysis and computational fluid dynamics (CFD). By leveraging advanced simulation techniques and statistical analysis, the research provides a robust framework for predicting mixture behavior under various operational conditions.

The proposed model demonstrated high predictive accuracy, as evidenced by the correlation coefficients (R^2) close to 1 for parameter μ and satisfactorily high for $\ln(\nu)$. This high level of accuracy indicates that the model is not only robust but also reliable for practical applications in the field of gas transportation and distribution. The use of dimensional analysis significantly reduced the computational complexity involved in the simulations, making the study more efficient. Focusing on key dimensionless numbers allowed the model to capture essential mixing dynamics without the need for extensive computational resources.

The integration of beta distribution functions to analyze mass fraction data further highlighted the ability of the model to handle complex data sets and provide meaningful insights into the mixing processes. These statistical tools helped simplify the data analysis, allowing for clearer interpretations of the mixing quality across different pipeline sections.

The successful application of the developed methodology and its validation suggest its potential for broader applications. Future work will aim to expand the range of simulation scenarios, potentially incorporating more diverse geometrical and operational conditions to further validate and refine the model. Additionally, exploring alternative algebraic expressions through symbolic regression, such as PySR [43], could offer new insights and enhance model predictability. While not explicitly analyzed in this paper, such enhancements could serve to address any unexplored dynamics or to improve the precision of the model predictions in future iterations.

Table 3
Values of parameters A_1 to D_6 for the correlation equation for $\ln(\nu)$ for different L/d_{CH_4} values.

L/d_{CH_4}	1	2	5	10	20	50
A_1	−7.50E+3	−1.68E+4	−2.90E+4	−3.41E+4	−3.24E+4	−5.02E+4
A_2	1.19E+3	2.15E+3	−2.29E+3	−1.10E+4	−1.39E+4	−1.09E+4
A_3	3.07E+3	5.11E+3	9.75E+3	9.75E+3	1.19E+4	1.55E+4
A_4	6.39E+3	1.43E+4	2.51E+4	3.09E+4	2.86E+4	4.24E+4
A_5	−4.78E+3	−8.05E+3	−9.63E+3	−6.64E+3	−7.47E+2	−5.43E+3
A_6	1.01E+2	3.69E+2	−8.30E+2	−1.20E+3	−2.25E+3	−6.49E+3
A_7	−9.36E+2	−2.38E+3	−4.07E+3	−5.04E+3	−4.97E+3	−6.64E+3
A_8	−2.61E+2	−1.36E+3	−3.77E+3	−6.25E+3	−7.21E+3	−7.91E+3
A_9	2.24E+2	4.48E+2	8.79E+2	9.30E+2	1.03E+3	1.25E+3
A_{10}	−2.65E+3	−5.65E+3	−9.72E+3	−1.08E+4	−9.70E+3	−1.73E+4
A_{11}	−6.84E+0	−2.11E+2	−2.86E+3	−7.48E+3	−8.60E+3	−7.84E+3
A_{12}	1.51E+3	2.26E+3	4.17E+3	3.97E+3	5.18E+3	7.14E+3
B_1	3.77E+3	5.16E+3	8.02E+3	7.77E+3	5.51E+3	7.03E+3
B_2	−2.31E+3	−3.01E+3	−5.98E+3	−8.38E+3	−6.41E+3	−3.98E+3
B_3	−1.19E+3	−2.69E+3	−3.05E+3	−1.50E+2	−3.30E+3	−5.24E+3
B_4	−1.54E+3	−2.65E+3	−4.26E+3	−4.68E+3	−3.88E+3	−4.57E+3
B_5	7.47E+2	1.44E+3	2.46E+3	3.12E+3	2.76E+3	1.63E+3
B_6	−1.86E+2	−2.98E+2	−7.09E+2	−1.14E+3	−1.73E+3	−2.07E+3
B_7	8.86E+1	4.69E+1	−2.97E+2	−1.15E+3	−7.20E+2	−3.98E+2
B_8	−4.18E+1	−4.58E+2	−4.66E+2	−2.72E+2	−6.31E+2	−4.41E+2
B_9	4.07E+2	8.76E+2	1.17E+3	8.87E+2	1.51E+3	1.81E+3
C_1	−2.86E+1	−8.15E+2	−8.85E+2	−6.56E+2	−1.18E+3	−6.76E+2
C_2	7.49E+2	1.36E+3	1.57E+3	9.17E+2	2.11E+3	2.82E+3
C_3	1.41E+3	2.55E+3	4.33E+3	5.43E+3	4.84E+3	3.05E+3
C_4	−1.87E+3	−2.38E+3	−3.49E+3	−2.79E+3	−1.04E+3	−1.58E+3
C_5	1.79E+3	5.52E+3	1.15E+4	1.68E+4	1.87E+4	2.22E+4
C_6	−5.86E+1	1.49E+2	7.10E+1	−9.44E+2	−2.84E+2	2.46E+2
D_1	4.98E+2	1.03E+3	1.70E+3	2.01E+3	1.83E+3	2.07E+3
D_2	8.20E+1	1.42E+2	3.48E+2	5.47E+2	8.18E+2	9.78E+2
D_3	1.62E+1	1.17E+2	1.14E+2	5.51E+1	1.61E+2	1.26E+2
D_4	−1.78E+2	−3.60E+2	−6.25E+2	−7.96E+2	−7.01E+2	−3.91E+2
D_5	−6.95E+1	3.88E+0	2.06E+2	5.09E+2	3.81E+2	2.72E+2
D_6	−1.03E+2	−2.42E+2	−3.40E+2	−2.80E+2	−4.32E+2	−4.98E+2

This research contributes significantly to the field by offering a practical yet sophisticated approach to modeling gas mixtures in pipeline networks, which is crucial for the ongoing integration of hydrogen into existing natural gas infrastructures as part of broader energy transition efforts.

CRediT authorship contribution statement

Carlos Montañés: Conceptualization, Methodology, Supervision, Writing – original draft, Writing – review & editing. **Leyre Pardo:** Investigation, Data curation, Writing – original draft. **Jaime Milla-Val:** Formal analysis, Data curation. **Antonio Gómez:** Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT to improve the readability of the text. The authors reviewed any suggested wording and edited it as needed; they take full responsibility for the content of the publication.

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